Trinity College Dublin
Coláiste na Tríonóide, Baile Átha Cliath
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# Disentangled Accent Modeling for Neural Voice Conversion

## Accent Recognition and Speech Transformation

**Focus:** Machine Learning & Neural Speech Processing

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# PROJECT MOTIVATION & OBJECTIVES

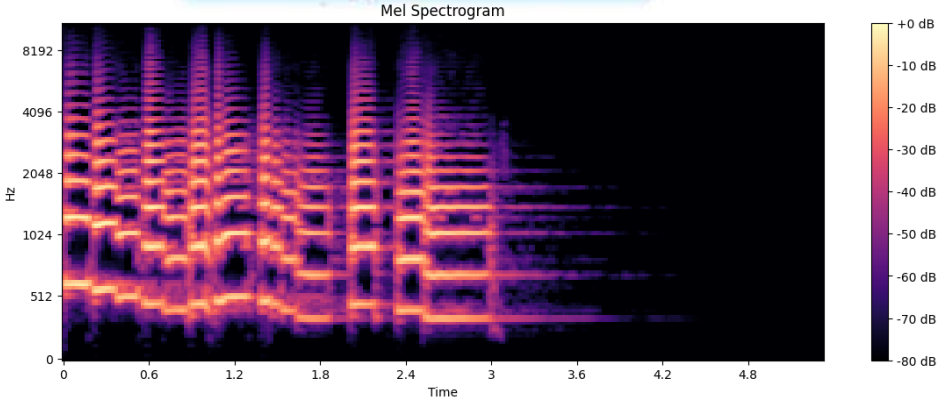
## Motivation

- **Accent variability** significantly degrades the performance and robustness of modern **speech AI systems**
- **Accent classification** identifies accent characteristics but does not resolve **accent mismatch** between speakers and systems
- **Neural voice conversion** enables controllable accent transformation through **disentangled accent representations**

## Key Objectives

- Develop a robust **accent classification** pipeline using acoustic features and **deep neural network models**
- Design an **accent-conditioned neural voice conversion framework** to transform speech between different accents
- Learn **disentangled accent representations** enabling controllable accent manipulation while preserving **speech content and speaker identity**

** Dialects of English in UK and Ireland*




*Feature Extraction: Mel-spectrogram Representation*

# Project Overview and System Framework

## Input

Speech Signal with Unknown Source Accent

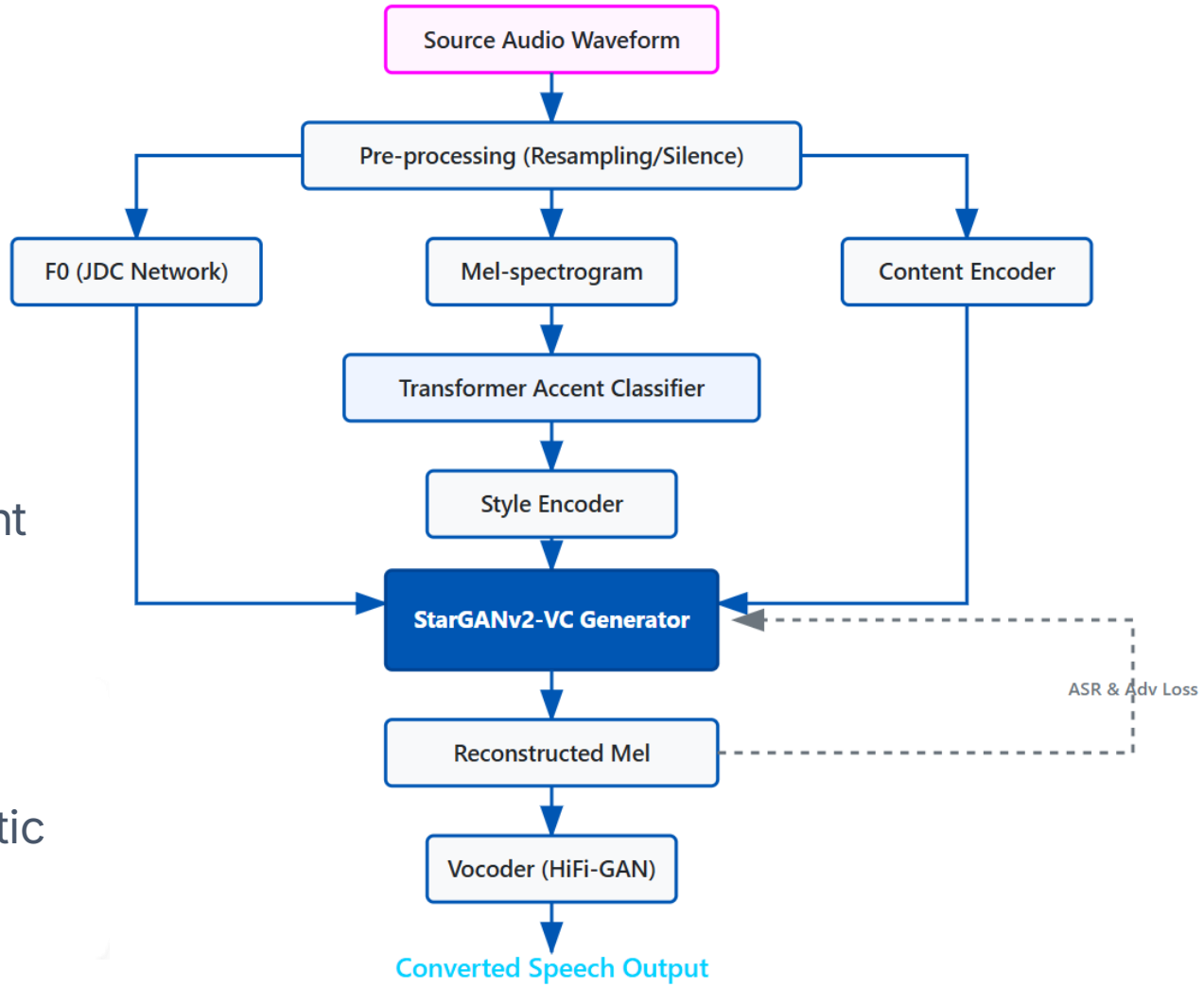
## Process

**Recognize:** Source Accent using accent classification model
**Condition:** Extract accent representation for conversion control
**Convert:** Transform speech into target accent using neural voice conversion model

## Output

Accent-converted speech preserving linguistic content and speaker identity

### System Architecture: Disentangled Accent Modeling



# Methodology

**Classifier → Target Accent → StarGANv2-VC conditioning**
"Accent classification provides explicit target selection, enabling controllable accent conversion."

## Accent Classification

- **Model:** Transformer-based classifier
- **Input:** Mel-spectrogram (log-mel scale)
- **Output:** Accent label (e.g., US / UK / IN / AU)
- **Goal:** Provide explicit target-accent control for the conversion stage

## Voice / Accent Conversion (StarGANv2-VC)

- **Framework:** StarGANv2-VC (non-parallel, many-to-many)
- **Key idea:** **Content preservation** + **speaker/style transfer** with **F0-consistent** conversion
- **How:** Style encoder / mapping network → style code; JDC F0 features → pitch guidance; ASR perceptual loss → content consistency
- **Result:** Convert speech towards a chosen target accent while preserving linguistic content and speaker identity (as much as possible)

# Database

Two complementary datasets are used: Common Voice for accent classification and VCTK for voice conversion training.

## Mozilla Common Voice 24.0 (English)

Used for accent classification training

- **Scale: 3,779 hours** speech from **99,289 speakers** (**2.56M clips**)
- **Accent labeling:** Includes accent, gender, age, and locale metadata
- **Speech richness:** 1.7M sentences across diverse linguistic domains
- **Speaker diversity:** Large global volunteer speaker population

**Source: Mozilla Common Voice Dataset v24.0 (2025)*

## VCTK Corpus (University of Edinburgh)

Used for neural voice conversion training

- **Scale: 110 speakers, ~400 utterances** per speaker (~**44 hours** total speech)
- **Speaker diversity:** Multi-speaker corpus with distinct speaker identities
- **Accent diversity:** Diverse regional English accents
- **Recording quality:** Studio-quality recordings in controlled acoustic environment

*Source: University of Edinburgh VCTK Corpus (2019)*